

Preliminary Global Logistic Regression

VIF Multicollinearity Check + Full-Sample LR — No Segmentation | WRDS Data (Q1 2010 – Q4 2024)

1. Sample Overview

Parameter	Value
Data Sources	Compustat Quarterly Fundamentals + CRSP Monthly Returns (WRDS)
Benchmark	S&P 500 Return — crsp.msi (sprtn) from WRDS
Companies	500 S&P 500 constituents
Observations (after merge & dropna)	11,187
Date Range	Q1 2010 — Q4 2024 (15 years, 60 quarters)
Outperformers (Y = 1)	6,068 (54.2%)
Underperformers (Y = 0)	5,119 (45.8%)
Predictors	13 financial ratios computed from raw Compustat quarterly items

2. Variance Inflation Factor (VIF) — Multicollinearity Check

VIF measures how much each predictor's variance is inflated by correlation with other predictors. $VIF < 5$ is acceptable; $VIF \geq 10$ indicates problematic multicollinearity that inflates standard errors and reduces individual significance.

Predictor	VIF Score	Assessment
Return on Assets (ROA)	5.62	Moderate concern
Return on Equity (ROE)	1.89	Acceptable — no issue
Gross Profit Margin	2.85	Acceptable — no issue
Operating Margin	3.58	Acceptable — no issue
Net Profit Margin	6.46	Moderate concern
Asset Turnover	2.03	Acceptable — no issue
Current Ratio	1.15	Acceptable — no issue
Debt-to-Equity Ratio	1.69	Acceptable — no issue
R&D Intensity	1.94	Acceptable — no issue
Revenue Growth (QoQ)	1.11	Acceptable — no issue
Net Income Growth (QoQ)	1.27	Acceptable — no issue
Price-to-Earnings (P/E)	1.03	Acceptable — no issue
Book-to-Market Ratio	1.21	Acceptable — no issue

3. Global Model Fit — Is the Model Statistically Significant?

Metric	Value	Benchmark	Verdict
Likelihood Ratio p-value	0.0000	< 0.05 required	SIGNIFICANT
Likelihood Ratio Statistic	148.33	Higher = better fit	—
McFadden's Pseudo R ²	0.0096	> 0.20 acceptable	Very Weak
AUC (ROC Curve)	0.5590	> 0.70 strong	Near-Random
Sample Size (N)	11,187	—	Adequate

4. Individual Predictor Coefficients & p-Values

Predictor	Coefficient	Std Error	Z-Stat	p-Value	Significant?
Return on Assets (ROA)	+0.0068	0.0456	0.149	0.8816	No
Return on Equity (ROE)	+0.0115	0.0264	0.437	0.6623	No
Gross Profit Margin	-0.0249	0.0323	-0.771	0.4410	No
Operating Margin	-0.0499	0.0362	-1.378	0.1681	No
Net Profit Margin	+0.0526	0.0489	1.075	0.2822	No
Asset Turnover	+0.0238	0.0273	0.873	0.3828	No
Current Ratio	+0.0810	0.0207	3.913	0.0001	*** p<0.01
Debt-to-Equity Ratio	+0.0113	0.0250	0.451	0.6519	No
R&D Intensity	-0.0330	0.0266	-1.240	0.2150	No
Revenue Growth (QoQ)	+0.0472	0.0202	2.336	0.0195	** p<0.05
Net Income Growth (QoQ)	-0.0220	0.0216	-1.020	0.3079	No
Price-to-Earnings (P/E)	+0.0506	0.0195	2.597	0.0094	*** p<0.01
Book-to-Market Ratio	-0.2079	0.0215	-9.653	0.0000	*** p<0.01

Green rows = significant at 5% level. Yellow = marginal (10%).

5. Interpretation

The global logistic regression model is statistically significant overall (LR $p = 0.0000 < 0.05$), meaning the 13 financial ratios together do carry some detectable relationship with outperformance. However, the model's practical predictive power is extremely weak: an AUC of 0.5590 is barely above random chance (0.50), and a Pseudo R² of 0.0096 means the ratios explain less than 1.0% of the variation in outperformance. In other words, the model is statistically detectable but practically useless as a predictor at the whole-market level.

This result is expected and is the central motivation for the sector-specific analysis. Pooling all 500 companies ignores the fact that the same ratio means different things across industries — a high Debt-to-Equity is normal for utilities but alarming for tech. Mixing these contradictory signals cancels predictive content at the aggregate level. Sector-specific models are expected to restore significance in 4–6 of the 11 GICS sectors.

Data: Compustat Quarterly Fundamentals + CRSP Monthly Returns via WRDS (lanagidan9790). Benchmark: S&P; 500 Composite Return from crsp.msi. Period: Q1 2010 – Q4 2024.